

Bachelor's Thesis Assignment



170297

Institut: Department of Computer Graphics and Multimedia (DCGM)
Student: **Kočí Otakar**
Programme: Information Technology
Title: **Procedural generator of 3D voxel maps**
Category: Computer Graphics
Academic year: 2025/26

Assignment:

1. Study methods of game map procedural generation with focus on vegetation, water, and other structures.
2. Propose an procedural generator of 3D voxel game maps in form of game demo.
3. Implement the proposed solution.
4. Evaluate the results.
5. Create a demonstration video.

Literature:

- Hendrikx, M., Meijer, S., Van Der Velden, J., & Iosup, A. (2013). Procedural content generation for games: A survey. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, 9(1), 1-22.
- Short, T., & Adams, T. (Eds.). (2017). *Procedural generation in game design*. CRC Press.
- Ratul, R., Sultana, S., Tasnim, J., & Rahman, A. (2019, December). Applicability of space colonization algorithm for real time tree generation. In *2019 22nd International Conference on Computer and Information Technology (ICCIT)* (pp. 1-6). IEEE.
- Gènevaux, J. D., Galin, É., Guérin, E., Peytavie, A., & Benes, B. (2013). Terrain generation using procedural models based on hydrology. *ACM Transactions on Graphics (TOG)*, 32(4), 1-13.
- Haji, S. H., & Abdulazeez, A. M. (2021). Comparison of optimization techniques based on gradient descent algorithm: A review. *PalArch's Journal of Archaeology of Egypt/Egyptology*, 18(4), 2715-2743.

Requirements for the semestral defence:

Points 1, 2 and the prototype of point 3.

Detailed formal requirements can be found at <https://www.fit.vut.cz/study/theses/>

Supervisor: **Vlnas Michal, Ing.**
Head of Department: Černocký Jan, prof. Dr. Ing.
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